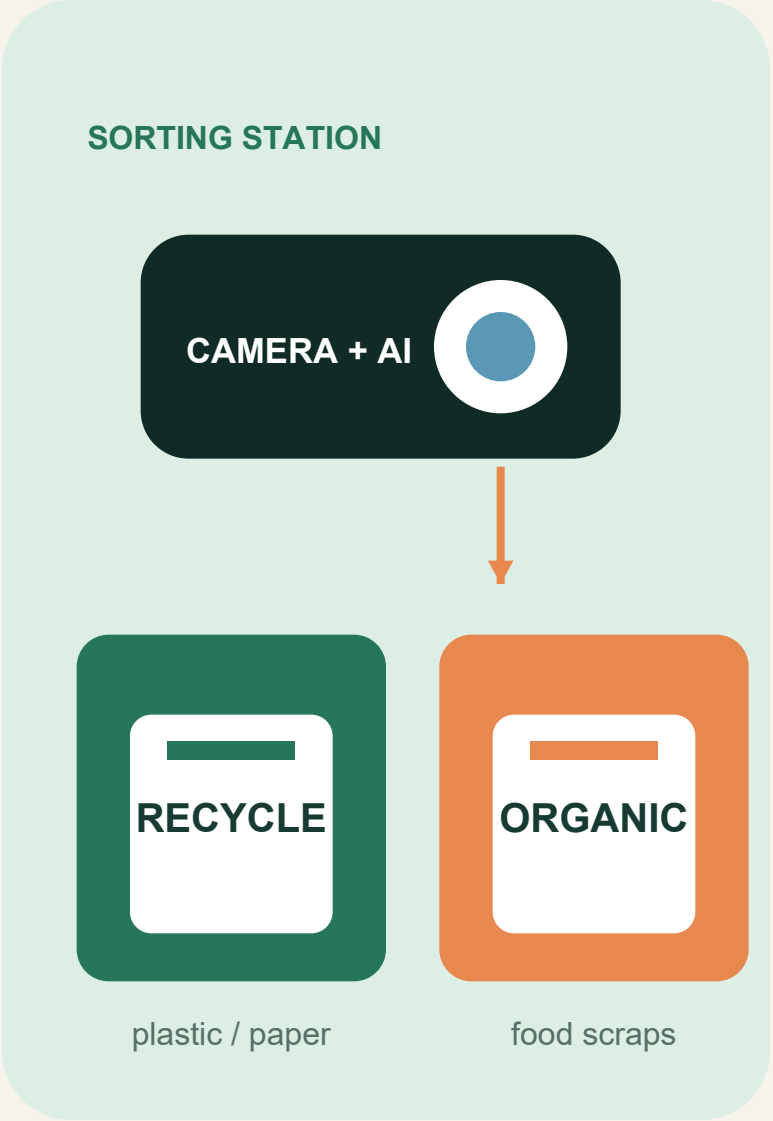
HACKATHON | AI / ML

SortWise AI

A camera AI that identifies waste and guides people to the right bin.

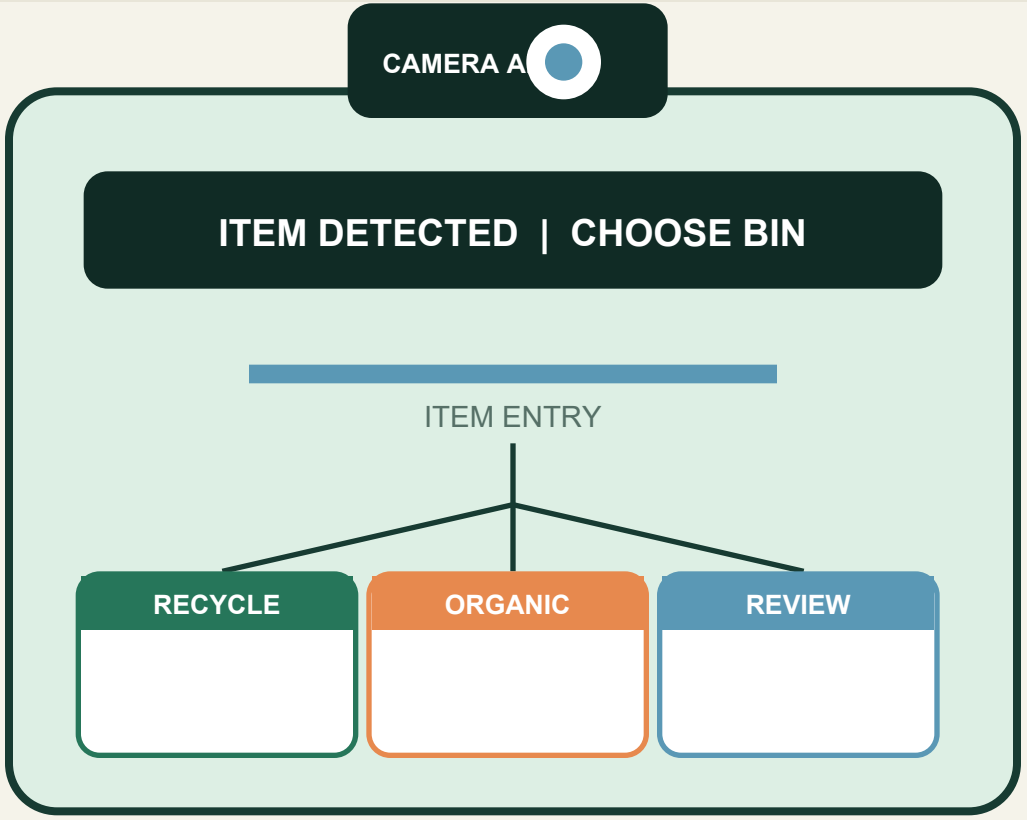
SEE IT → IDENTIFY IT → SORT IT

Smart waste segregation assistant



Presented by Team Akari

A camera-guided sorting station



SORTING IN PRACTICE

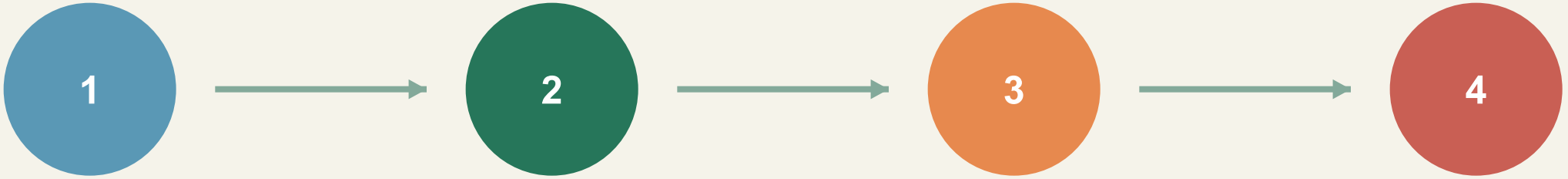


SortWise adds camera guidance at the point where people discard an item.

RECYCLABLE ORGANIC E-WASTE HAZARDOUS GENERAL

Photo courtesy USEPA | Public domain via Wikimedia Commons

Camera identifies waste and recommends a bin



CAMERA

Captures the item

AI MODEL

Identifies the item

LOCAL RULES

Select the right route

BIN GUIDANCE

Screen, light or flap

LOW CONFIDENCE? Ask the user to confirm or request staff help.

Suspected batteries get special collection guidance.

The model identifies an item; local rules determine the disposal route.

Mixed waste makes recovery harder

2.56B tonnes

Global municipal waste in 2022

2050 projection: 3.86B tonnes

COSTS OF POOR SEGREGATION

01

Recycling contamination

Food residue can make clean materials harder to process.

02

Methane emissions

Food waste can release methane as it decomposes in landfills.

03

Safety risk

Damaged batteries in ordinary waste can create fire risks.

Sources: World Bank, What a Waste 3.0 (2026); UNEP, waste and methane; US EPA, lithium-ion batteries.

A clear process for the user

1. **Place one item in front of the camera**
2. AI identifies the item
3. Screen shows the recommended bin
4. If uncertain, ask the user or staff to confirm

WASTE CATEGORIES

- RECYCLABLE
- ORGANIC
- E-WASTE
- HAZARDOUS
- GENERAL

Follow the local waste disposal rules.

MVP: a small prototype with a complete workflow

WHAT THE DEMO SHOWS

- Live camera feed with detection box
- Bottle, paper and food waste examples
- Safe fallback for uncertain items
- Bin indicator; optional servo flap
- Dashboard with sorting counts

WHAT WE MEASURE

Precision and recall by class

Prediction latency

Unknown / review rate

User correction rate

Prove impact with a real site pilot.

SortWise AI identifies items and guides people to the right bin.

Prototype scope: selected items in a controlled camera setup. Use cautious handling for batteries.

THANK YOU

TEAM AKARI

Questions &
Discussion

Thank you for listening.